

Direct Bound for Coreset Forecasting (MAE, Finite/Discrete)

1 Notation

Datasets and models.

$$D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n, \quad D_{\text{test}} = \{(x_j, y_j)\}_{j=1}^m, \quad D_{\text{coreset}} \subset D_{\text{train}}, \quad |D_{\text{coreset}}| = k. \quad (1)$$

$$f_{\theta}^{(D_{\text{train}})} := \text{model trained on } D_{\text{train}}, \quad f_{\theta}^{(D_{\text{coreset}})} := \text{model trained on } D_{\text{coreset}}. \quad (2)$$

MAE on a finite set.

$$\text{MAE}_D(f) := \frac{1}{|D|} \sum_{(x,y) \in D} |f(x) - y|. \quad (3)$$

Input sets and ground distance.

$$X_D := \{x \mid (x, y) \in D\}, \quad d_{ST}(x, x') : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}. \quad (4)$$

Wasserstein-1 on finite/discrete sets. For

$$X_A = \{a_i\}_{i=1}^{n_A}, \quad X_B = \{b_j\}_{j=1}^{n_B}, \quad (5)$$

$$W_1^{d_{ST}}(X_A, X_B) := \min_{\Gamma \in \Pi(n_A, n_B)} \sum_{i=1}^{n_A} \sum_{j=1}^{n_B} \Gamma_{ij} d_{ST}(a_i, b_j), \quad (6)$$

where

$$\Pi(n_A, n_B) := \left\{ \Gamma \in \mathbb{R}_{\geq 0}^{n_A \times n_B} \mid \sum_j \Gamma_{ij} = \frac{1}{n_A}, \sum_i \Gamma_{ij} = \frac{1}{n_B} \right\}. \quad (7)$$

2 Direct Derivation (Readable Version)

Primary target term (what we want to bound).

$$G_{\text{test}} := \left| \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right|. \quad (8)$$

Signed gap (used only for “who is better” statement).

$$\Delta_{\text{test}} := \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) - \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{train}})} \right). \quad (9)$$

Kantorovich–Rubinstein inequality (finite/discrete, fixed model). Assume each input on the compared support has a unique target value $y(x)$. For a fixed model f , define

$$\phi_f(x) := |f(x) - y(x)|. \quad (10)$$

If ϕ_f is L_f -Lipschitz with respect to d_{ST} , then

$$\left| \frac{1}{n_A} \sum_{i=1}^{n_A} \phi_f(a_i) - \frac{1}{n_B} \sum_{j=1}^{n_B} \phi_f(b_j) \right| \leq L_f W_1^{d_{ST}}(X_A, X_B). \quad (11)$$

Hence,

$$|\text{MAE}_A(f) - \text{MAE}_B(f)| \leq L_f W_1^{d_{ST}}(X_A, X_B). \quad (12)$$

Intuition: with a fixed model, average MAE changes mainly when the input set itself moves far in Wasserstein distance.

Step 1: decompose the primary target term G_{test} .

$$\begin{aligned} G_{\text{test}} \leq & \underbrace{\left| \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{train}})} \right) \right|}_{\text{(A) test-train shift for train model}} \\ & + \underbrace{\left| \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right|}_{\text{(B) train-set model gap}} \\ & + \underbrace{\left| \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) - \text{MAE}_{D_{\text{test}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right|}_{\text{(C) train-test shift for coreset model}}. \end{aligned} \quad (13)$$

Step 2: apply KR to (A) and (C).

$$\text{(A)} \leq L_{\text{train}} W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}), \quad \text{(C)} \leq L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}). \quad (14)$$

So,

$$G_{\text{test}} \leq (L_{\text{train}} + L_{\text{core}}) W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}) + \text{(B)}. \quad (15)$$

Step 3: bound (B) using train-coreset mismatch. Add and subtract coreset MAE for each model:

$$\begin{aligned} \text{(B)} \leq & \left| \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{train}})} \right) \right| \\ & + \left| \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right| \\ & + \left| \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) - \text{MAE}_{D_{\text{train}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right|. \end{aligned} \quad (16)$$

Define

$$\varepsilon_{\text{opt}} := \left| \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{train}})} \right) - \text{MAE}_{D_{\text{coreset}}} \left(f_{\theta}^{(D_{\text{coreset}})} \right) \right|. \quad (17)$$

Applying KR to the first and third terms gives

$$\text{(B)} \leq (L_{\text{train}} + L_{\text{core}}) W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}}) + \varepsilon_{\text{opt}}. \quad (18)$$

Final bound on the primary target term.

$$\begin{aligned} G_{\text{test}} \leq & (L_{\text{train}} + L_{\text{core}}) W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}) \\ & + (L_{\text{train}} + L_{\text{core}}) W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}}) + \varepsilon_{\text{opt}}. \end{aligned} \quad (19)$$

Interpretation.

- $W_1(X_{D_{\text{test}}}, X_{D_{\text{train}}})$: test-train shift (fixed by data split; coreset method cannot control this).
- $W_1(X_{D_{\text{train}}}, X_{D_{\text{coreset}}})$: train-coreset mismatch (directly controllable by selection).
- ε_{opt} : optimization gap on the coreset (training/protocol quality).
- $L_{\text{core}}, L_{\text{train}}$: sensitivity of each fixed model to input movement.

3 k-medoids Link

Let $X_{\text{train}} = \{x_i\}_{i=1}^n$ and choose medoids $C \subset X_{\text{train}}$, $|C| = k$. k-medoids minimizes

$$J_{\text{kmed}}(C) := \frac{1}{n} \sum_{i=1}^n \min_{c \in C} d_{ST}(x_i, c). \quad (20)$$

Define nearest-medoid assignment

$$\pi_C(x_i) := \arg \min_{c \in C} d_{ST}(x_i, c), \quad \pi_C(X_{\text{train}}) := \{\pi_C(x_i)\}_{i=1}^n. \quad (21)$$

The induced transport by nearest assignment gives

$$W_1^{d_{ST}}(X_{\text{train}}, \pi_C(X_{\text{train}})) \leq J_{\text{kmed}}(C). \quad (22)$$

So minimizing J_{kmed} directly shrinks a Wasserstein mismatch term. This is exactly aligned with the controllable term $W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}})$ in the bound above.

4 When Coreset Can Beat Train Model

Target signed term.

$$\Delta_{\text{test}} := \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{coreset}})}) - \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{train}})}). \quad (23)$$

We want to show when $\Delta_{\text{test}} < 0$, i.e., when the coreset-trained model is better on test.

Step 1: one-sided upper bound for the coreset model on test. From the same KR + triangle steps used above:

$$\begin{aligned} \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{coreset}})}) &\leq \text{MAE}_{D_{\text{coreset}}}(f_{\theta}^{(D_{\text{coreset}})}) \\ &\quad + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}) \\ &\quad + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}}). \end{aligned} \quad (24)$$

Define the right-hand side as

$$U_{\text{core}} := \text{MAE}_{D_{\text{coreset}}}(f_{\theta}^{(D_{\text{coreset}})}) + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}) + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}}). \quad (25)$$

So,

$$\text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{coreset}})}) \leq U_{\text{core}}. \quad (26)$$

Step 2: convert to a bound on the target term Δ_{test} . Subtract $\text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{train}})})$ from both sides:

$$\Delta_{\text{test}} \leq U_{\text{core}} - \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{train}})}). \quad (27)$$

Sufficient condition for coreset win.

$$U_{\text{core}} < \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{train}})}) \implies \Delta_{\text{test}} < 0. \quad (28)$$

Equivalent expanded form:

$$\begin{aligned} &\text{MAE}_{D_{\text{coreset}}}(f_{\theta}^{(D_{\text{coreset}})}) + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{test}}}, X_{D_{\text{train}}}) \\ &\quad + L_{\text{core}} W_1^{d_{ST}}(X_{D_{\text{train}}}, X_{D_{\text{coreset}}}) < \text{MAE}_{D_{\text{test}}}(f_{\theta}^{(D_{\text{train}})}). \end{aligned} \quad (29)$$

This is a sufficient condition, not a necessary condition.